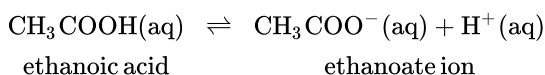


26.1 The acidity of carboxylic acids

Carboxylic acids display the typical reactions of all acids due to the presence of excess $\text{H}^+(\text{aq})$ ions in their aqueous solutions (see [Section 21.3](#)). For example, they react with bases to form a salt and water. Their salts are called *carboxylates*. The carboxylate salt formed by the reaction of ethanoic acid with sodium hydroxide is called sodium ethanoate, $\text{CH}_3\text{COO}^-\text{Na}^+$.

The carboxylic acids are weak acids. The majority of their molecules are undissociated in water.

For example:



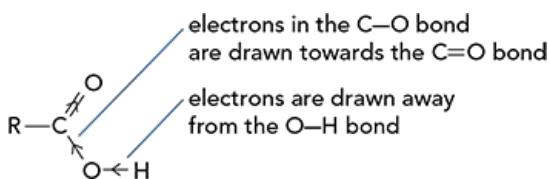
The position of this equilibrium lies well over to the left-hand side. The dissociation constant, K_a , of ethanoic acid at 25 °C is $1.7 \times 10^{-5} \text{ mol dm}^{-3}$ (see [Section 21.3](#)).

IMPORTANT

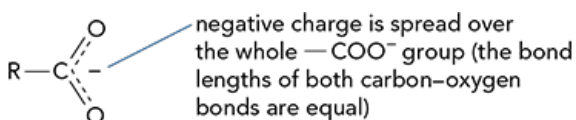
Remember that the smaller the value of K_a , the weaker the acid.

The carboxylic acids are stronger acids than alcohols. This is explained by:

- the O–H bond in the carboxylic acid is weakened by the carbonyl group, $\text{C}=\text{O}$.



- The carboxylate ion is stabilised by the delocalisation of electrons around the —COO^- group. This delocalisation spreads out the negative charge on the carboxylate ion, reducing its charge density. This makes it less likely to bond with an $\text{H}^+(\text{aq})$ ion to re-form the undissociated acid molecule with its —COOH group.



As we have already seen in [Section 25.3](#), phenol ($\text{C}_6\text{H}_5\text{OH}$) is a stronger acid than ethanol. However, phenol ($K_a = 1.0 \times 10^{-10} \text{ mol dm}^{-3}$) is a weaker acid than ethanoic acid, CH_3COOH ($K_a = 1.7 \times 10^{-5} \text{ mol dm}^{-3}$) and benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$ ($K_a = 6.3 \times 10^{-5} \text{ mol dm}^{-3}$).

IMPORTANT

Carboxylic acids are stronger acids than phenols, and phenols are stronger acids than alcohols.

We find that electron-withdrawing groups bonded to the carbon atom next to the —COOH group make the acid stronger.

There are two reasons for this:

- electron-withdrawing groups further weaken the O–H bond in the undissociated acid molecule
- electron-withdrawing groups extend the delocalisation of the negative charge on the —COO^- group of the carboxylate ion, further increasing the stabilisation of the —COO^- group, and making it less likely to bond with an $\text{H}^+(\text{aq})$ ion.

Chlorine atoms are an example of an electron-withdrawing group. The dissociation constants (K_a) of

ethanoic acid and its three chloro-substituted derivatives are shown in Table 26.2.

Acid	K_a at 25 °C / mol dm ⁻³
ethanoic acid, CH ₃ COOH	1.7×10^{-5}
chloroethanoic acid, CH ₂ ClCOOH	1.3×10^{-3}
dichloroethanoic acid, CHCl ₂ COOH	5.0×10^{-2}
trichloroethanoic acid, CCl ₃ COOH	2.3×10^{-1}

Table 26.2: The larger the value of K_a , the stronger the acid.

Trichloroethanoic acid, CCl₃COOH, has three strongly electronegative Cl atoms all withdrawing electrons from the –COOH group, weakening the O–H bond more than the other acids in Table 26.2. Once the O–H bond is broken, the resulting anion is also stabilised more effectively as the negative charge is further spread out by its three electron-withdrawing Cl atoms, making it less attractive to H⁺(aq) ions. This makes the CCl₃COOH the strongest of the acids listed in Table 26.2 as it has most Cl atoms (see Figure 26.3).

IMPORTANT

The more electron-withdrawing groups on the carbon atom bonded to the C atom in the –COOH group, the stronger the acid.

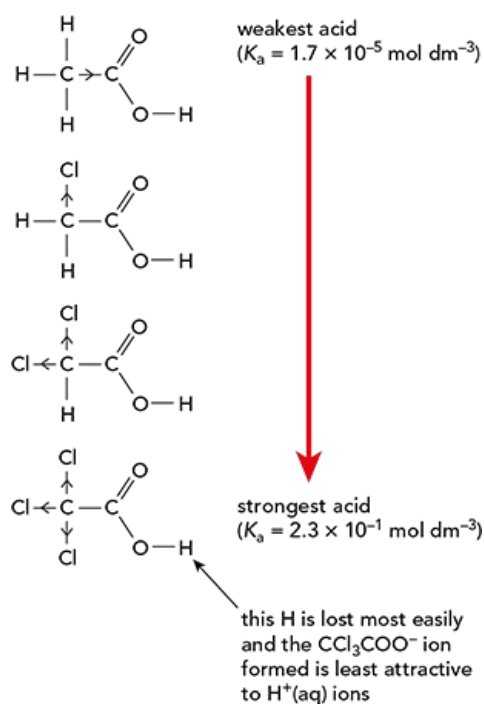


Figure 26.3: The trend in the acid strength of ethanoic acid and its chloro-substituted derivatives.

Ethanoic acid is the weakest acid in Table 26.2, as the methyl group is electron donating. This has the opposite effect to electron-withdrawing groups:

- it strengthens the O–H bond in the acid's –COOH group
- it donates negative charge towards the –COO⁻ group of the carboxylate ion, concentrating its negative charge, and making it more likely to accept an H⁺(aq) ion.

Question

Working in a pair or a small group, discuss Question 1 below together before writing down your answers.

- a** Place the following acids in order of strength, starting with the strongest acid.

$\text{CH}_3\text{CH}_2\text{COOH}$ $\text{CH}_3\text{CCl}_2\text{COOH}$ $\text{CH}_3\text{CHClCOOH}$

- b** Explain why ethanoic acid is a stronger acid than ethanol.
- c** Predict which would be the stronger acid: methanoic acid or ethanoic acid. Explain your reasoning.

26.2 Oxidation of two carboxylic acids

You saw in [Chapter 17](#) that primary alcohols can be oxidised by heating with acidified potassium dichromate(VI) solution to form aldehydes and then further oxidation produces carboxylic acids (see [Section 18.2](#)). The carboxylic acids prepared are not usually oxidised any further.

However, *methanoic acid* (HCOOH) is a stronger reducing agent than other carboxylic acids, so it can undergo further oxidation (Figure 26.4).

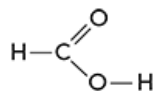
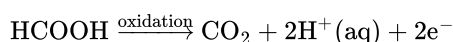


Figure 26.4: The displayed formula of methanoic acid.

This involves the oxidising agent breaking down the methanoic acid molecule, forming *carbon dioxide*. This oxidation can be carried out even by warming with mild oxidising agents, such as the Fehling's or Tollens' reagents used to distinguish between aldehydes and ketones (see [Section 18.5](#)).

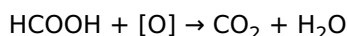
In a positive test for an aldehyde, the Cu^{2+} ion in Fehling's solution is reduced to the Cu^+ ion, which precipitates out as red copper(I) oxide. When methanoic acid is oxidised by Fehling's solution, the same positive test result is obtained. With Tollens' reagent, the silver ion present, Ag^+ , is reduced to silver metal, Ag, when it oxidises methanoic acid (also as in the positive test for an aldehyde).

The half-equation for the oxidation of methanoic acid can be written in terms of electron transfer to an oxidising agent as:



Oxidation no. +2 +4
of carbon:

or in terms of the addition of oxygen from an oxidising agent as:



This oxidation of methanoic acid will also occur with stronger oxidising agents such as acidified solutions of potassium manganate(VII) (decolourising the purple solution) or potassium dichromate(VI) (turning the orange solution green).

Another common example of a carboxylic acid that can be oxidised by these stronger oxidising agents is *ethanedioic acid* (a dicarboxylic acid). Its molecular formula is $\text{H}_2\text{C}_2\text{O}_4$. Its structural formula is HOOC-COOH or $(\text{COOH})_2$, and its displayed formula is shown in Figure 26.5.

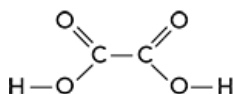
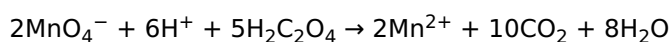


Figure 26.5: The displayed formula of ethanedioic acid.

A solution of ethanedioic acid (with sulfuric acid added) can be used in titrations to standardise a solution of potassium manganate(VII). As with methanoic acid, the oxidation results in the formation of carbon dioxide and water. The solid ethanedioic acid can be weighed out accurately to make a standard solution of known concentration. This is warmed in a conical flask and the potassium manganate(VII) solution to be standardised is run in from a burette. The end-point of the titration is identified when the addition of a drop of the purple manganate(VII) solution turns the solution pink. This happens when all the ethanedioic acid has been oxidised.

The balanced ionic equation for this redox reaction is:



The manganese(II) ions formed catalyse the reaction. This is an example of autocatalysis, in which one of the products of the reaction acts as a catalyst for that reaction.

Question

- 2 a** Carboxylic acids can be prepared by refluxing a primary alcohol with a strong oxidising agent. Why is methanoic acid never prepared by refluxing methanol with acidified potassium dichromate(VI) solution?
- b** Look back at the two half-equations for the oxidation of methanoic acid. Write two similar equations representing the oxidation of ethanedioic acid.
- c i** In a titration between ethanedioic acid and potassium manganate(VII) solutions, ethanedioic acid is heated in its conical flask before adding the potassium manganate(VII) solution from the burette. Suggest why.
- ii** In an experiment to standardise a solution of potassium manganate(VII), 25.0 cm³ of a standard solution of 0.0500 mol dm⁻³ ethanedioic acid was fully oxidised by a titre of 8.65 cm³ of potassium manganate(VII) solution. Calculate the concentration of the potassium manganate(VII) solution.

REFLECTION

This activity helps you to reflect on your understanding of redox reactions, half-equations and titration calculations.

Discuss with a partner whether or not you both feel that you have a good understanding of redox reactions, half-equations and titration calculations. Are you confident in all three areas or not? Think about the skills and thought processes you need to master these areas. This will help you identify where any problems lie.

If necessary, look back to [Chapters 1, 7 and 24](#) to revise redox reactions, half-equations and titration calculations. How will you use the skills and thought processes you discussed with your partner when revising these chapters?

26.3 Acyl chlorides

Preparing acyl chlorides

An **acyl chloride** is similar in structure to a carboxylic acid but the hydroxyl group, -OH , has been replaced by a Cl atom. The displayed formula of ethanoyl chloride is shown in Figure 26.6.

The structural formula of ethanoyl chloride is written as CH_3COCl .

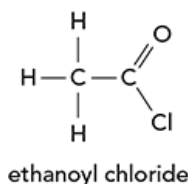
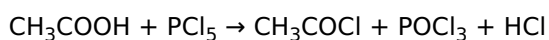


Figure 26.6: Ethanoyl chloride is the acyl chloride derived from ethanoic acid.

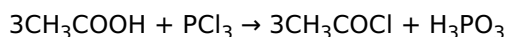
We can prepare acyl chlorides from their corresponding carboxylic acid using solid phosphorus(V) chloride, liquid phosphorus(III) chloride or liquid sulfur dichloride oxide (SOCl_2).

With phosphorus(V) chloride



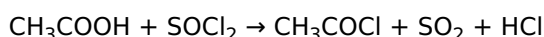
No special conditions are required for this reaction.

With phosphorus(III) chloride



Heat is required for this reaction to take place.

With sulfur dichloride oxide



No special conditions are required for this reaction.

You can observe steamy white fumes when hydrogen chloride gas given off. The third reaction, with SOCl_2 , is the one reaction to produce the acyl chloride as the *only* liquid product, as both SO_2 and HCl are gases.

Question

3 Write a balanced equation, using structural formulae for the organic compounds, and state any conditions necessary, to show the formation of:

- a** propanoyl chloride from a suitable carboxylic acid and SOCl_2
- b** methanoyl chloride from a suitable carboxylic acid and PCl_3
- c** butanoyl chloride from a suitable carboxylic acid and PCl_5 .

Check your answers to Question 3 against those of another learner, making sure all your equations are balanced, before looking at the answers provided.

Reactions of acyl chlorides

Many useful compounds can be synthesised from carboxylic acids. However, the synthetic reactions can be difficult to carry out because carboxylic acids are relatively unreactive. One way round this is to first convert the carboxylic acid into an acyl chloride. Acyl chlorides are *much* more reactive than carboxylic

acids.

As we saw in [Section 18.2](#), an acyl chloride is similar in structure to a carboxylic acid, with the -OH group replaced by a Cl atom. So the -COOH functional group becomes the -COCl group. The structural formula of acyl chlorides can be written as ROCl , where R is an alkyl or aryl group. For example, propanoyl chloride can be written as $\text{CH}_3\text{CH}_2\text{COCl}$.

The structural formulae of benzoic acid and its derivative benzoyl chloride are shown in Figure 26.7.

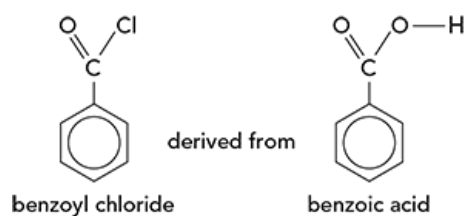
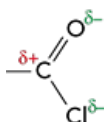


Figure 26.7: Benzoyl chloride is the acyl chloride derived from benzoic acid.

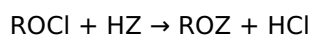
Acyl chlorides are reactive compounds. The carbonyl carbon has electrons drawn away from it by the Cl atom as well as by its O atom as both are strongly electronegative atoms. This gives the carbonyl carbon a relatively large partial positive charge and makes it particularly open to attack from nucleophiles. Remember that nucleophiles can donate a lone pair of electrons to an electron-deficient carbon atom in the mechanism of a reaction.



The acyl chlorides are reactive liquids. When they react with nucleophiles the C-Cl bond breaks and white fumes of hydrogen chloride, HCl , are given off.

IMPORTANT

To generalise the reactions of acyl chlorides, we can write:



where HZ can be water, an alcohol, phenols, ammonia or an amine.

Note that HZ contains either an oxygen or nitrogen atom with a lone pair of electrons that can be donated. HZ can therefore act as a nucleophile.

Also note that in the reactions with ammonia and amines, the HCl gas formed reacts quickly with the ammonia / amines to form salts. Ammonia will form NH_4^+Cl^- , primary amines form $\text{RNH}_3^+\text{Cl}^-$ and secondary amines form $\text{R}_2\text{NH}_2^+\text{Cl}^-$. This means that you see far less of the white HCl fumes given off than with water, alcohols or phenols.

Reaction with water

Figure 26.8 shows the *mechanism of hydrolysis*, showing the initial attack by a water molecule, acting as a nucleophile, followed by the elimination of a molecule of hydrogen chloride.

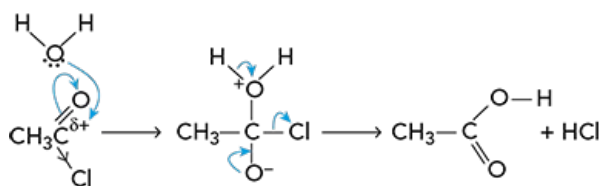


Figure 26.8: The condensation (addition-elimination) mechanism of hydrolysis of ethanoyl chloride.

The hydrolysis can be classified as a condensation reaction, as there is an initial *addition* reaction of water

across the C=O bond, followed by *elimination* of a small molecule, in this case HCl.

This reaction is far more vigorous than the hydrolysis of chloroalkanes (see [Section 16.2](#)). The hydrolysis of chloroalkanes needs a strong alkali, such as aqueous sodium hydroxide, to be refluxed with the chloroalkane to hydrolyse it. The nucleophile needed to hydrolyse a chloroalkane is the negatively charged hydroxide ion, OH⁻.

However, a neutral water molecule is sufficient to hydrolyse an acyl chloride quickly at room temperature. This difference is due to the carbon bonded to the chlorine atom in a chloroalkane not being as strongly $\delta+$ as the carbon atom in an acyl chloride. Remember that in an acyl chloride, the carbon bonded to the chlorine atom is also attached to an oxygen atom. So the carbonyl carbon has two strongly electronegative atoms pulling electrons away from it. Therefore, the attack by the nucleophile is much more rapid.

Aryl chlorides or chloroarenes, such as chlorobenzene, C₆H₅Cl, will *not* undergo hydrolysis. The carbon atom bonded directly to the chlorine atom is part of the delocalised π bonding system of the benzene ring. A lone pair of electrons from the Cl atom tends to overlap with the delocalised p electrons in the benzene ring. This causes the C–Cl bond to have some double bond character, making it stronger and more difficult to break, so hydrolysis does not occur.

IMPORTANT

The ease of hydrolysis, starting with the compounds most readily broken down, is:

acyl chloride > chloroalkane > aryl chloride or chloroarene

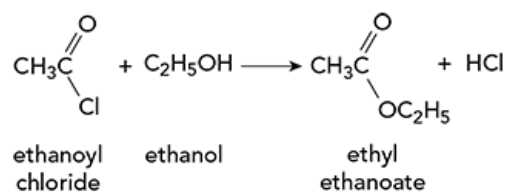
Question

- 4 **a** In its reaction with a nucleophile, explain why an acyl chloride reacts faster than an alcohol.
- b** Name the products of the hydrolysis of propanoyl chloride.
- c** **i** Place the following compounds in order of ease of hydrolysis, starting with the most reactive.
- C₆H₅Cl CH₃CH₂COCl CH₃CH₂CH₂Cl
- ii** Explain your answer to part **i**.
- iii** If a reaction occurs with water, what will you see in part **i**?

Reaction with alcohols and with phenol

When acyl chlorides react with alcohols and phenol they form esters (and HCl). The reactions happen more quickly than the reactions of alcohols or phenol with carboxylic acids. The acyl chloride reactions also go to completion and do not form an equilibrium mixture like the reactions with carboxylic acids. Therefore acyl chlorides are useful in the synthesis of esters in the chemical industry.

Ethanoyl chloride will react vigorously with ethanol to form an ester:



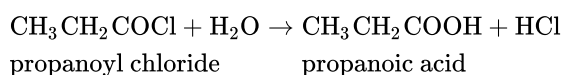
PRACTICAL ACTIVITY 26.1

Hydrolysis of acyl chlorides

Water is a suitable nucleophile to attack an acyl chloride molecule.

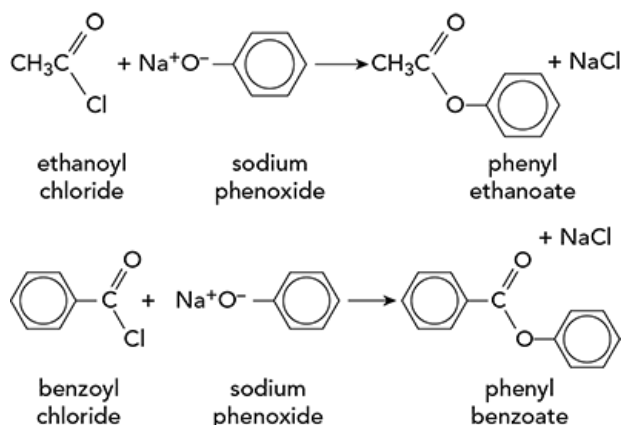
A lone pair of electrons on the oxygen atom in a water molecule initiates the attack on the $\delta+$ carbonyl carbon atom. The reaction produces a carboxylic acid and hydrogen chloride gas. It is an example of a hydrolysis reaction, i.e. the breakdown of a compound by water.

For example:



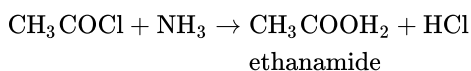
The acyl chloride can be added to water using a dropping pipette. The reaction is immediate and white fumes of HCl are observed rising from the liquid.

With phenol, the reaction with an acyl chloride proceeds if warmed. There is no reaction between phenol and carboxylic acids, so acyl chlorides must be used if you want to make phenyl esters. The reaction takes place in the presence of a base. The initial reaction between phenol and the base creates the phenoxide ion, $\text{C}_6\text{H}_5\text{O}^-$. The negatively charged phenoxide ion acts as the nucleophile to attack the carbonyl carbon in the acyl chloride:

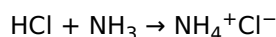


Reaction with ammonia and amines

Ammonia (NH_3) acts as the nucleophile in the usual nucleophilic addition-elimination reaction of acyl chlorides. The concentrated ammonia is added at room temperature and the reaction is vigorous:



followed by:



The organic product is an amide ($\text{R}-\text{CONH}_2$), in which the Cl atom in the acyl chloride is replaced by an $-\text{NH}_2$ group. The mechanism of the reaction with ammonia is shown in Figure 26.9:

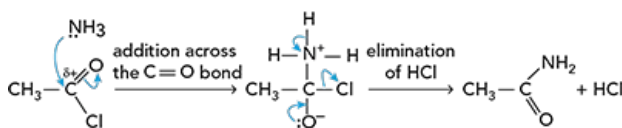
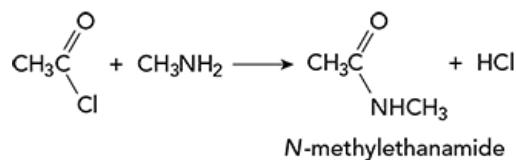


Figure 26.9: Mechanism of the nucleophilic addition-elimination of an acyl chloride with ammonia.

Primary amines ($\text{R}-\text{NH}_2$) can also act as nucleophiles as they contain nitrogen atoms with a lone pair of electrons. This lone pair of electrons is available to attack the carbonyl carbon atom in acyl chlorides.

The reaction of an acyl chloride with an amine is vigorous and the organic product is a substituted amide.

For example, with a primary amine such as methylamine:



The *mechanism* of the nucleophilic condensation (addition-elimination) reactions of acyl chlorides with

amines follows the same steps as the hydrolysis with water in Figure 26.8 earlier in this section, and the reaction with ammonia above (Figure 26.9). You can imagine the primary amine as an ammonia molecule with one of its H atoms replaced by an alkyl group.

Look at Figure 26.10.

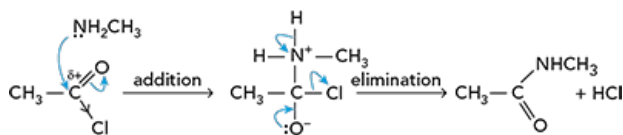
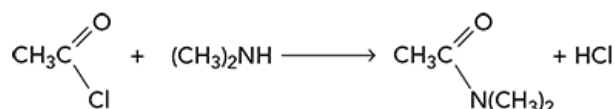


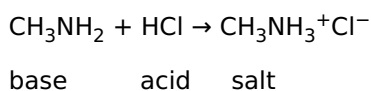
Figure 26.10: The nucleophilic condensation (addition-elimination) mechanism of ethanoyl chloride reacting with a primary amine.

With secondary amines ($\text{R}^1\text{R}^2\text{NH}$), such as dimethylamine, the reaction is similar:



In both the reactions with amines shown above, most of the HCl is not given off as white fumes. As with ammonia, most of the HCl undergoes an acid-base reaction with unreacted amine to form a white organic ammonium salt. With methylamine, we get $\text{CH}_3\text{NH}_3^+\text{Cl}^-$, and with dimethylamine $(\text{CH}_3)_2\text{NH}_2^+\text{Cl}^-$ is formed.

For example:



Question

- 5 a** Using an acyl chloride as a starting compound in each case, name the reactants you would use to make:
- ethyl ethanoate
 - methyl butanoate
 - phenyl benzoate.
- b** Copy and complete the following equation:
- $$\text{CH}_3\text{CH}_2\text{COCl} + \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2 \rightarrow \text{_____} + \text{_____}$$

REFLECTION

- Working with a partner, discuss the mechanism of nucleophilic addition-elimination in the reaction between ethanoyl chloride and dimethylamine. Then try showing the mechanism using curly arrows.
- Now discuss which nucleophile you would expect to be most effective: ammonia, a primary amine or a secondary amine. Think about the inductive effect of alkyl groups.
- Talk together about how you go about drawing mechanisms and how they help to make sense of organic reactions.
- Discuss if there are any parts of [Chapter 26](#) that you need to:
 - read through again to make sure you really understand
 - seek more guidance on, even after going over it again.

Use the checklist in the Summary to help you reflect.

SUMMARY

Carboxylic acids are weak acids. Their strength is increased if the carbon atom next to the —COOH group has electron-withdrawing atoms, such as chlorine, bonded to it.

Acyl chlorides (RCOCl) are made by reacting carboxylic acids with PCl_5 , PCl_3 or SOCl_2 .

Acyl chlorides react with amines to give substituted amides. Acyl chlorides are more reactive than their corresponding carboxylic acids.

Acyl chlorides are reactive liquids. They are easily hydrolysed by water, forming a carboxylic acid while giving off white fumes of hydrogen chloride.

Acyl chlorides also react with alcohols and phenols to give esters.

The ease of hydrolysis of chloro compounds is largely determined by the size of the partial positive charge on the carbon in the C—Cl bond and its strength. So acyl chlorides are more reactive with water than alkyl chlorides (halogenoalkanes), which are more reactive than the hydrolysis-resistant aryl chlorides (halogenoarenes).

EXAM-STYLE QUESTIONS

- 1** Acyl chlorides and carboxylic acids can both be used to prepare esters.
- a**
- i** Give the reagents and conditions required to make ethyl ethanoate directly from a carboxylic acid. **[3]**
 - ii** Write an equation to show the formation of ethyl ethanoate in part **a i**. **[1]**
- b**
- i** Give the reagents and conditions required to make phenyl benzoate from an acyl chloride. **[3]**
 - ii** Write an equation to show the formation of phenyl benzoate in part **b i**. **[1]**
 - iii** Describe what you see happen in this reaction. **[1]**
- c** Describe two advantages that acyl chlorides have over carboxylic acids in the preparation of esters in industry. **[2]**
- [Total: 11]**
- 2**
- a** Draw the displayed formula of butanoyl chloride. **[1]**
- b**
- i** Give the name and formula of a reagent that can be used to prepare butanoyl chloride from butanoic acid. **[2]**
 - ii** Write a balanced equation for the reaction in part **b i**. **[1]**
 - iii** Butanoic acid is a weaker acid than 2-chlorobutanoic acid. Explain why. **[2]**
 - iv** Draw the skeletal formula of 2-chlorobutanoic acid. **[1]**
 - v** Name a chloro-substituted butanoic acid that is a stronger acid than 2-chlorobutanoic acid. **[1]**
 - vi** Explain the relative ease of hydrolysis for the compounds chlorobenzene, butanoyl chloride, and 1-chlorobutane. **[4]**
- [Total: 12]**

SELF-EVALUATION

After studying this chapter, complete a table like this:

I can	See section...	Needs more work	Almost there	Ready to move on
explain the relative acidity of carboxylic acids and of chlorine-substituted ethanoic acids	26.1			
describe how some carboxylic acids, such as methanoic acid and ethanedioic acid, can be further oxidised	26.2			
describe the reactions of carboxylic acids in the preparation of acyl chlorides	26.3			
describe the hydrolysis of acyl chlorides	26.3			
describe the reactions of acyl chlorides with alcohols, phenols, ammonia and primary or secondary amines	26.3			
describe the condensation (addition-elimination) mechanism for the reactions of acyl chlorides	26.3			
explain the relative ease of hydrolysis of acyl chlorides, alkyl chlorides and halogenoarenes (aryl chlorides).	26.3			